

```
In [3]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
data = pd.read_excel("data.xlsx")
```

```
In [4]: print(data.head())
```

	Restaurant ID	Restaurant Name	Country Code	City \
0	7402935	Skye	94	Jakarta
1	7410290	Satoo - Hotel Shangri-La	94	Jakarta
2	7420899	Sushi Masa	94	Jakarta
3	7421967	3 Wise Monkeys	94	Jakarta
4	7422489	Avec Moi Restaurant and Bar	94	Jakarta

	Address \
0	Menara BCA, Lantai 56, Jl. MH. Thamrin, Thamri...
1	Hotel Shangri-La, Jl. Jend. Sudirman
2	Jl. Tuna Raya No. 5, Penjaringan
3	Jl. Suryo No. 26, Senopati, Jakarta
4	Gedung PIC, Jl. Teluk Betung 43, Thamrin, Jakarta

	Locality	Locality Verbose \
0	Grand Indonesia Mall, Thamrin	Grand Indonesia Mall, Thamrin, Jakarta
1	Hotel Shangri-La, Sudirman	Hotel Shangri-La, Sudirman, Jakarta
2	Penjaringan	Penjaringan, Jakarta
3	Senopati	Senopati, Jakarta
4	Thamrin	Thamrin, Jakarta

	Longitude	Latitude	Cuisines	Average Cost for two \
0	106.821999	-6.196778	Italian, Continental	800000
1	106.818961	-6.203292	Asian, Indonesian, Western	800000
2	106.800144	-6.101298	Sushi, Japanese	500000
3	106.813400	-6.235241	Japanese	450000
4	106.821023	-6.196270	French, Western	350000

	Currency	Has Table booking	Has Online delivery	Price range \
0	Indonesian Rupiah(IDR)	No	No	3
1	Indonesian Rupiah(IDR)	No	No	3
2	Indonesian Rupiah(IDR)	No	No	3
3	Indonesian Rupiah(IDR)	No	No	3
4	Indonesian Rupiah(IDR)	No	No	3

	Aggregate rating	Rating color	Rating text	Votes
0	4.1	Green	Very Good	1498
1	4.6	Dark Green	Excellent	873
2	4.9	Dark Green	Excellent	605
3	4.2	Green	Very Good	395
4	4.3	Green	Very Good	243

```
In [5]: print(data.tail())
```

	Restaurant ID	Restaurant Name	Country Code	City \
9546	18279289	BMG - All Day Dining	1	Dehradun
9547	2300497	Atmosphere Grill Cafe Sheesha	1	Kanpur
9548	18312106	UrbanCrave	1	Kanpur
9549	3900245	Deena Chat Bhandar	1	Varanasi
9550	18246202	VNS Live Studio	1	Varanasi

	Address	Locality \
9546	140 A, Rajpur Road, Jakhan, Dehradun	Jakhan
9547	8th Floor, J.S. Tower, 16/106 - Mall Road, Kan...	Mall Road
9548	14/125, The Mall, Mall Road, Colonelganj, Para...	Parade
9549	D-47/184, Luxa Road, Dashaswmedh Road, Varanasi	Dashaswmedh Road
9550	Hotel Varuna Ground Floor, 22 Gulab Bagh, Sigr...	Sigra

	Locality Verbose	Longitude	Latitude \
9546	Jakhan, Dehradun	78.068890	30.362686
9547	Mall Road, Kanpur	80.354002	26.472001
9548	Parade, Kanpur	80.342796	26.474986
9549	Dashaswmedh Road, Varanasi	0.000000	0.000000
9550	Sigra, Varanasi	82.991694	25.318345

	Cuisines	Average Cost for two \
9546	Chinese, North Indian, Fast Food	0
9547	Indian, Chinese, Continental	0
9548	Cafe, Continental, Desserts, Ice Cream, Italia...	0
9549	Street Food	0
9550	Chinese, North Indian	0

	Currency	Has Table booking	Has Online delivery	Price range \
9546	Indian Rupees(Rs.)	No	No	1
9547	Indian Rupees(Rs.)	No	No	1
9548	Indian Rupees(Rs.)	No	No	1
9549	Indian Rupees(Rs.)	No	No	1
9550	Indian Rupees(Rs.)	No	No	1

	Aggregate rating	Rating color	Rating text	Votes
9546	4.3	Green	Very Good	63
9547	3.6	Yellow	Good	34
9548	3.9	Yellow	Good	127
9549	3.8	Yellow	Good	78
9550	3.5	Yellow	Good	109

In [6]: `print(data.info())`

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9551 entries, 0 to 9550
Data columns (total 19 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Restaurant ID                        9551 non-null   int64
1   Restaurant Name                      9550 non-null   object
2   Country Code                        9551 non-null   int64
3   City                                9551 non-null   object
4   Address                             9551 non-null   object
5   Locality                            9551 non-null   object
6   Locality Verbose                    9551 non-null   object
7   Longitude                           9551 non-null   float64
8   Latitude                           9551 non-null   float64
9   Cuisines                            9542 non-null   object
10  Average Cost for two                 9551 non-null   int64
11  Currency                            9551 non-null   object
12  Has Table booking                   9551 non-null   object
13  Has Online delivery                 9551 non-null   object
14  Price range                         9551 non-null   int64
15  Aggregate rating                    9551 non-null   float64
16  Rating color                        9551 non-null   object
17  Rating text                         9551 non-null   object
18  Votes                               9551 non-null   int64
dtypes: float64(3), int64(5), object(11)
memory usage: 1.4+ MB
None

```

```
In [8]: print(data.shape)
```

```
(9551, 19)
```

```
In [9]: print(data.describe)
```

	<bound method NDFrame.describe of ame Country Code	City \	Restaurant ID	Restaurant N
0	7402935		Skye	94 Jakarta
1	7410290	Satoo - Hotel Shangri-La		94 Jakarta
2	7420899		Sushi Masa	94 Jakarta
3	7421967		3 Wise Monkeys	94 Jakarta
4	7422489	Avec Moi Restaurant and Bar		94 Jakarta
...	...		...	...
9546	18279289	BMG - All Day Dining		1 Dehradun
9547	2300497	Atmosphere Grill Cafe Sheesha		1 Kanpur
9548	18312106		UrbanCrave	1 Kanpur
9549	3900245	Deena Chat Bhandar		1 Varanasi
9550	18246202	VNS Live Studio		1 Varanasi

	Address \
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2	Jl. Tuna Raya No. 5, Penjaringan
3	Jl. Suryo No. 26, Senopati, Jakarta
4	Gedung PIC, Jl. Teluk Betung 43, Thamrin, Jakarta
...	...
9546	140 A, Rajpur Road, Jakhan, Dehradun
9547	8th Floor, J.S. Tower, 16/106 - Mall Road, Kan...
9548	14/125, The Mall, Mall Road, Colonelganj, Para...
9549	D-47/184, Luxa Road, Dashaswmedh Road, Varanasi
9550	Hotel Varuna Ground Floor, 22 Gulab Bagh, Sigr...

	Locality	Locality Verbose \
0	Grand Indonesia Mall, Thamrin	Grand Indonesia Mall, Thamrin, Jakarta
1	Hotel Shangri-La, Sudirman	Hotel Shangri-La, Sudirman, Jakarta
2	Penjaringan	Penjaringan, Jakarta
3	Senopati	Senopati, Jakarta
4	Thamrin	Thamrin, Jakarta
...	...	...
9546	Jakhan	Jakhan, Dehradun
9547	Mall Road	Mall Road, Kanpur
9548	Parade	Parade, Kanpur
9549	Dashaswmedh Road	Dashaswmedh Road, Varanasi
9550	Sigra	Sigra, Varanasi

	Longitude	Latitude \
0	106.821999	-6.196778
1	106.818961	-6.203292
2	106.800144	-6.101298
3	106.813400	-6.235241
4	106.821023	-6.196270
...	...	...
9546	78.068890	30.362686
9547	80.354002	26.472001
9548	80.342796	26.474986
9549	0.000000	0.000000
9550	82.991694	25.318345

	Cuisines	Average Cost for two \
0	Italian, Continental	800000
1	Asian, Indonesian, Western	800000
2	Sushi, Japanese	500000
3	Japanese	450000
4	French, Western	350000
...	...	...

9546	Chinese, North Indian, Fast Food	0
9547	Indian, Chinese, Continental	0
9548	Cafe, Continental, Desserts, Ice Cream, Italia...	0
9549	Street Food	0
9550	Chinese, North Indian	0

	Currency	Has Table booking	Has Online delivery	\
0	Indonesian Rupiah(IDR)	No	No	
1	Indonesian Rupiah(IDR)	No	No	
2	Indonesian Rupiah(IDR)	No	No	
3	Indonesian Rupiah(IDR)	No	No	
4	Indonesian Rupiah(IDR)	No	No	
...	...	...	...	
9546	Indian Rupees(Rs.)	No	No	
9547	Indian Rupees(Rs.)	No	No	
9548	Indian Rupees(Rs.)	No	No	
9549	Indian Rupees(Rs.)	No	No	
9550	Indian Rupees(Rs.)	No	No	

	Price range	Aggregate rating	Rating color	Rating text	Votes
0	3	4.1	Green	Very Good	1498
1	3	4.6	Dark Green	Excellent	873
2	3	4.9	Dark Green	Excellent	605
3	3	4.2	Green	Very Good	395
4	3	4.3	Green	Very Good	243
...	...	...	...	...	...
9546	1	4.3	Green	Very Good	63
9547	1	3.6	Yellow	Good	34
9548	1	3.9	Yellow	Good	127
9549	1	3.8	Yellow	Good	78
9550	1	3.5	Yellow	Good	109

[9551 rows x 19 columns]>

```
In [10]: print(data.isnull().sum())
```

```
Restaurant ID      0
Restaurant Name    1
Country Code       0
City               0
Address            0
Locality           0
Locality Verbose   0
Longitude          0
Latitude           0
Cuisines           9
Average Cost for two 0
Currency           0
Has Table booking  0
Has Online delivery 0
Price range        0
Aggregate rating    0
Rating color        0
Rating text         0
Votes              0
dtype: int64
```

```
In [11]: print(data.duplicated().sum())
```

0

```
In [12]: data = data.drop_duplicates()
```

```
In [13]: print(data.duplicated().sum())
```

0

```
In [14]: print(data.shape)
```

(9551, 19)

```
In [18]: Country_data = pd.read_excel("Country-Code.xlsx")
```

```
In [20]: print(Country_data.head())
```

	Country Code	Country
0	1	India
1	14	Australia
2	30	Brazil
3	37	Canada
4	94	Indonesia

```
In [21]: print(Country_data.shape)
```

(15, 2)

```
In [24]: print(Country_data.isnull().sum())
```

Country Code 0
Country 0
dtype: int64

```
In [29]: data=data.merge(Country_data,on="Country Code",how="left")
```

```
In [31]: print(data.columns)
```

Index(['Restaurant ID', 'Restaurant Name', 'Country Code', 'City', 'Address',
'Locality', 'Locality Verbose', 'Longitude', 'Latitude', 'Cuisines',
'Average Cost for two', 'Currency', 'Has Table booking',
'Has Online delivery', 'Price range', 'Aggregate rating',
'Rating color', 'Rating text', 'Votes', 'Country'],
 dtype='object')

```
In [34]: city_counts = data ['City'].value_counts()
```

```
In [35]: city_counts
```

```
Out[35]: City
New Delhi      5473
Gurgaon        1118
Noida          1080
Faridabad       251
Ghaziabad       25
...
Vernonia        1
Dicky Beach     1
Inverloch       1
Lakes Entrance  1
Miller          1
Name: count, Length: 141, dtype: int64
```

```
In [36]: max_count = city_counts.max()
```

```
In [39]: print("Number of restaurants:",max_count)
```

Number of restaurants: 5473

```
In [40]: max_city = city_counts.idxmax()
```

```
In [41]: print("city with maximum restaurants:",max_city)
```

city with maximum restaurants: New Delhi

```
In [9]: import pandas as pd  
data = pd.read_excel("data.xlsx")
```

```
In [10]: city_counts = data ['City'].value_counts()
```

```
In [12]: city_counts
```

```
Out[12]: City  
New Delhi      5473  
Gurgaon        1118  
Noida          1080  
Faridabad       251  
Ghaziabad       25  
...  
Vernonia        1  
Dicky Beach     1  
Inverloch       1  
Lakes Entrance  1  
Miller          1  
Name: count, Length: 141, dtype: int64
```

```
In [13]: min_count =city_counts.idxmin()
```

```
In [15]: print("City with minimum restaurants",min_count)
```

City with minimum restaurants Bandung

```
In [16]: min_count = city_counts.min()
```

```
In [17]: print("Number of restaurants:", min_count)
```

Number of restaurants: 1

```
In [18]: franchise_presence = data.groupby("Restaurant Name")["City"].nunique()
```

```
In [20]: franchise_presence.idxmax()
```

```
Out[20]: 'Barbeque Nation'
```

```
In [21]: city_counts = data['City'].value_counts()
```

```
In [22]: city_counts
```

```
Out[22]: City
New Delhi      5473
Gurgaon        1118
Noida          1080
Faridabad       251
Ghaziabad       25
...
Vernonia        1
Dicky Beach     1
Inverloch       1
Lakes Entrance  1
Miller          1
Name: count, Length: 141, dtype: int64
```

```
In [23]: data.columns
```

```
Out[23]: Index(['Restaurant ID', 'Restaurant Name', 'Country Code', 'City', 'Address',
               'Locality', 'Locality Verbose', 'Longitude', 'Latitude', 'Cuisines',
               'Average Cost for two', 'Currency', 'Has Table booking',
               'Has Online delivery', 'Price range', 'Aggregate rating',
               'Rating color', 'Rating text', 'Votes'],
              dtype='object')
```

```
In [42]: franchise_presence = data.groupby("Restaurant Name")["Country Code"].nunique().r
```

```
In [43]: franchise_presence
```

```
Out[43]:
```

	Restaurant Name	Country Code
0	12212	1
1	Let's Burrp	1
2	#45	1
3	#Dilliwaala6	1
4	#InstaFreeze	1
...	...	...
7440	t Lounge by Dilmah	1
7441	tashas	1
7442	wagamama	1
7443	{Niche} - Cafe & Bar	1
7444	làukura€Ùa Sofras€±	1

7445 rows × 2 columns

```
In [34]: print(type(franchise_presence))
print(franchise_presence.head())
```



```
<class 'pandas.core.frame.DataFrame'>
  Restaurant Name  City
0          12212    1
1    Let's Burrp    1
2           #45    1
3    #Dilliwaala6    1
4    #InstaFreeze    1
```

```
In [38]: type(franchise_presence)
```

```
Out[38]: pandas.core.frame.DataFrame
```

```
In [49]: franchise_presence = data.groupby("Restaurant Name")["Country Code"].nunique()
```

```
In [50]: top = franchise_presence.sort_values(ascending=False).head(10)
print(top)
```

```
Restaurant Name
Hard Rock Cafe      3
Indian Coffee House 2
Ichiban             2
Gazebo              2
Jamie's Italian     2
Din Tai Fung        2
Barbeque Nation     2
Gymkhana            2
Chili's Grill & Bar  2
Applebee's          2
Name: Country Code, dtype: int64
```

```
In [52]: table_booking_counts = data["Has Table booking"].value_counts()
print(table_booking_counts)
```

```
Has Table booking
No      8393
Yes     1158
Name: count, dtype: int64
```

```
In [57]: table_ratio = table_booking_counts/table_booking_counts.sum()
print(table_ratio)
```

```
Has Table booking
No      0.878756
Yes     0.121244
Name: count, dtype: float64
```

```
In [58]: table_ratio = (table_booking_counts / table_booking_counts.sum()) * 100
print(table_ratio)
```

```
Has Table booking
No      87.875615
Yes     12.124385
Name: count, dtype: float64
```

```
In [66]: import pandas as pd
data = pd.read_excel("data.xlsx")
```

```
In [70]: Online_delivery = data['Has Online delivery'].value_counts()
print(Online_delivery)
```

```
Has Online delivery
No      7100
Yes     2451
Name: count, dtype: int64
```

```
In [80]: percentage = data['Has Online delivery'].str.strip().str.lower().eq('yes').sum()
```

```
In [81]: print(f"Percentage of restaurants providing online delivery: {percentage:.2f}%")
```

Percentage of restaurants providing online delivery: 25.66%

```
In [5]: import pandas as pd
data = pd.read_excel("data.xlsx")
```

```
In [13]: votes_deliver = data[data["Has Online delivery"] == True]["Votes"].sum()
votes_not_deliver = data[data["Has Online delivery"] == False]["Votes"].sum()
difference = votes_deliver - votes_not_deliver
print(difference)
```

0

```
In [14]: df_cuisines = data.assign(Cuisines=data["Cuisines"].str.split(", ").explode("Cu
```

```
In [15]: cuisine_counts = df_cuisines["Cuisines"].value_counts()
```

```
In [16]: top_10_cuisines = cuisine_counts.head(10)
```

```
In [17]: print(top_10_cuisines)
```

```
Cuisines
North Indian    3960
Chinese         2735
Fast Food       1986
Mughlai         995
Italian         764
Bakery          745
Continental     736
Cafe            703
Desserts        653
South Indian    636
Name: count, dtype: int64
```

```
In [19]: print(data.columns)
```

```
Index(['Restaurant ID', 'Restaurant Name', 'Country Code', 'City', 'Address',
       'Locality', 'Locality Verbose', 'Longitude', 'Latitude', 'Cuisines',
       'Average Cost for two', 'Currency', 'Has Table booking',
       'Has Online delivery', 'Price range', 'Aggregate rating',
       'Rating color', 'Rating text', 'Votes'],
      dtype='object')
```

```
In [20]: data["cuisine_count"] = data["Cuisines"].fillna("").apply(
        lambda x: len([c for c in x.split(",") if c.strip() != ""]))
```

```
In [21]: max_cuisines = data["cuisine_count"].max()
min_cuisines = data["cuisine_count"].min()
```

```
In [22]: print("Maximum number of cuisines served by a restaurant:", max_cuisines)
print("Minimum number of cuisines served by a restaurant:", min_cuisines)
```

Maximum number of cuisines served by a restaurant: 8
Minimum number of cuisines served by a restaurant: 0

```
In [25]: df_cuisine_city = data.copy()
df_cuisine_city["Cuisines"] = df_cuisine_city["Cuisines"].fillna("").str.split(")
```

```
In [26]: df_cuisine_city = df_cuisine_city.explode("Cuisines")
df_cuisine_city["Cuisines"] = df_cuisine_city["Cuisines"].str.strip()
```

```
In [27]: df_cuisine_city = df_cuisine_city[df_cuisine_city["Cuisines"] != ""]
```

```
In [28]: city_cuisine_counts = (
    df_cuisine_city.groupby(["City", "Cuisines"])
    .size()
    .reset_index(name="Count")
)
```

```
In [29]: most_served_per_city = (
    city_cuisine_counts
    .sort_values(["City", "Count"], ascending=[True, False])
    .groupby("City")
    .head(1)
    .reset_index(drop=True)
)

print(most_served_per_city)
```

	City	Cuisines	Count
0	Abu Dhabi	Indian	7
1	Agra	North Indian	15
2	Ahmedabad	Continental	12
3	Albany	American	4
4	Allahabad	North Indian	12
..	...	...	...
135	Weirton	Burger	1
136	Wellington City	Cafe	9
137	Winchester Bay	Burger	1
138	Yorkton	Asian	1
139	İstanbul	Cafe	4

[140 rows x 3 columns]

```
In [30]: data["Average Cost for two"].describe()
```

```
Out[30]: count    9551.000000
mean      1199.210763
std       16121.183073
min         0.000000
25%        250.000000
50%        400.000000
75%        700.000000
max       800000.000000
Name: Average Cost for two, dtype: float64
```

```
In [31]: cost_distribution = data["Average Cost for two"].value_counts().sort_index()
print(cost_distribution)
```

```

Average Cost for two
0          18
7           4
10         128
15          4
20          25
...
300000      2
350000      1
450000      1
500000      1
800000      2
Name: count, Length: 140, dtype: int64

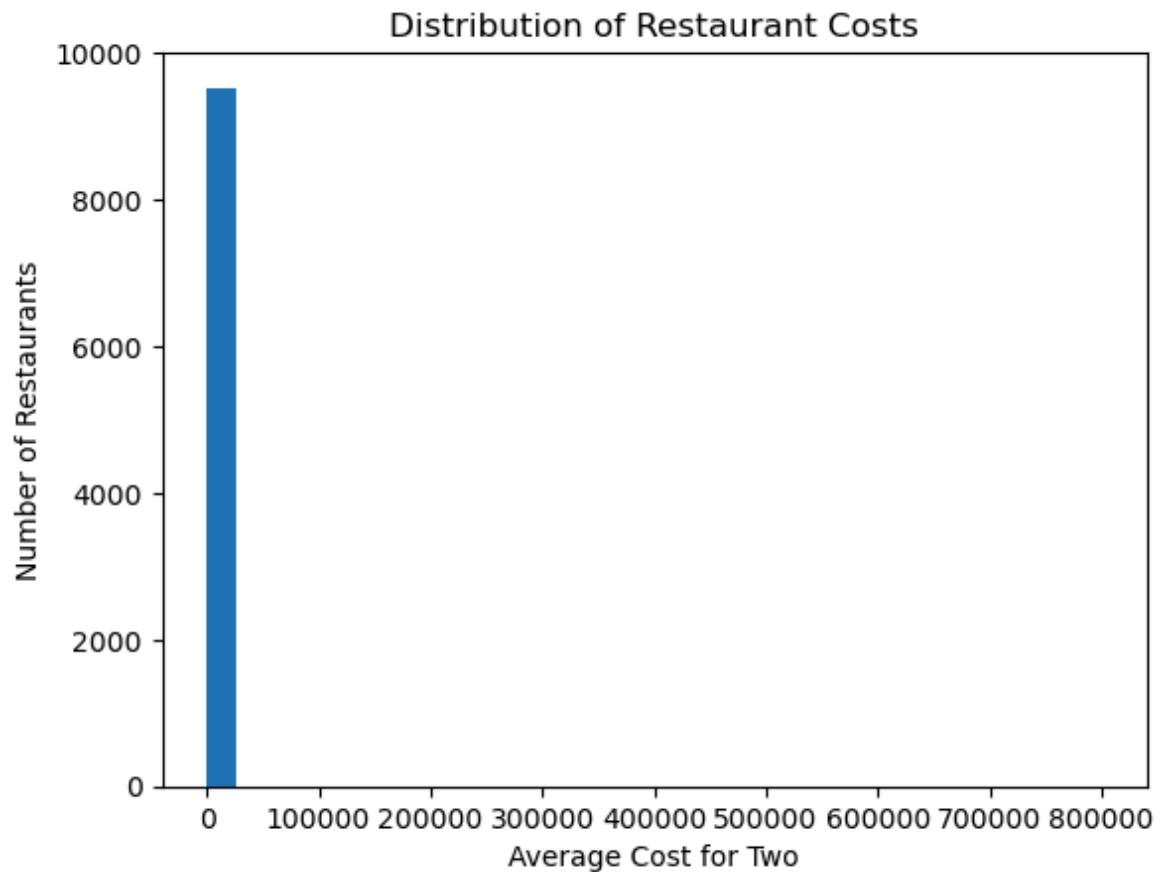
```

```

In [32]: import matplotlib.pyplot as plt

plt.hist(data["Average Cost for two"], bins=30)
plt.xlabel("Average Cost for Two")
plt.ylabel("Number of Restaurants")
plt.title("Distribution of Restaurant Costs")
plt.show()

```



```

In [33]: city_cost_dist = (
    data.groupby("City")["Average Cost for two"]
    .describe()
)

print(city_cost_dist)

```

	count	mean	std	min	25%	50%	\
City							
Abu Dhabi	20.0	182.000000	117.455479	50.0	95.00	165.0	
Agra	20.0	1065.000000	934.443262	0.0	537.50	825.0	
Ahmedabad	21.0	857.142857	369.555718	300.0	600.00	900.0	
Albany	20.0	19.750000	17.507517	0.0	10.00	10.0	
Allahabad	20.0	517.500000	215.989157	0.0	400.00	500.0	
...	...	...	...	...	...	...	
Weirton	1.0	25.000000	NaN	25.0	25.00	25.0	
Wellington City	20.0	71.250000	38.212529	25.0	50.00	57.5	
Winchester Bay	1.0	25.000000	NaN	25.0	25.00	25.0	
Yorkton	1.0	25.000000	NaN	25.0	25.00	25.0	
İstanbul	14.0	81.428571	39.779613	30.0	51.25	80.0	

	75%	max
City		
Abu Dhabi	250.00	500.0
Agra	1200.00	3600.0
Ahmedabad	1000.00	1800.0
Albany	25.00	70.0
Allahabad	600.00	1000.0
...	...	...
Weirton	25.00	25.0
Wellington City	82.50	200.0
Winchester Bay	25.00	25.0
Yorkton	25.00	25.0
İstanbul	101.25	170.0

[141 rows x 8 columns]

```
In [34]: data["Aggregate rating"].value_counts().sort_index()
```

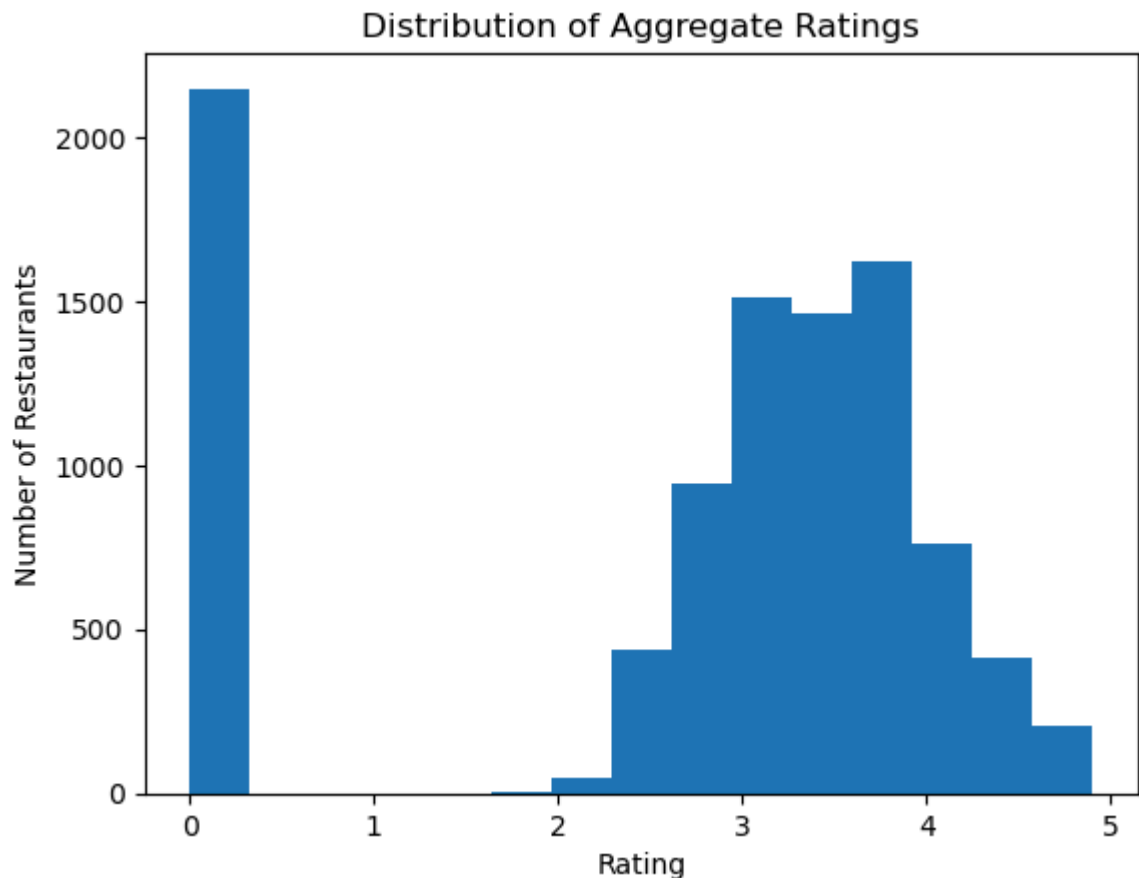
Out[34]: Aggregate rating

0.0	2148
1.8	1
1.9	2
2.0	7
2.1	15
2.2	27
2.3	47
2.4	87
2.5	110
2.6	191
2.7	250
2.8	315
2.9	381
3.0	468
3.1	519
3.2	522
3.3	483
3.4	498
3.5	480
3.6	458
3.7	427
3.8	400
3.9	335
4.0	266
4.1	274
4.2	221
4.3	174
4.4	144
4.5	95
4.6	78
4.7	42
4.8	25
4.9	61

Name: count, dtype: int64

In [35]: `import matplotlib.pyplot as plt`

```
plt.hist(data["Aggregate rating"], bins=15)
plt.xlabel("Rating")
plt.ylabel("Number of Restaurants")
plt.title("Distribution of Aggregate Ratings")
plt.show()
```



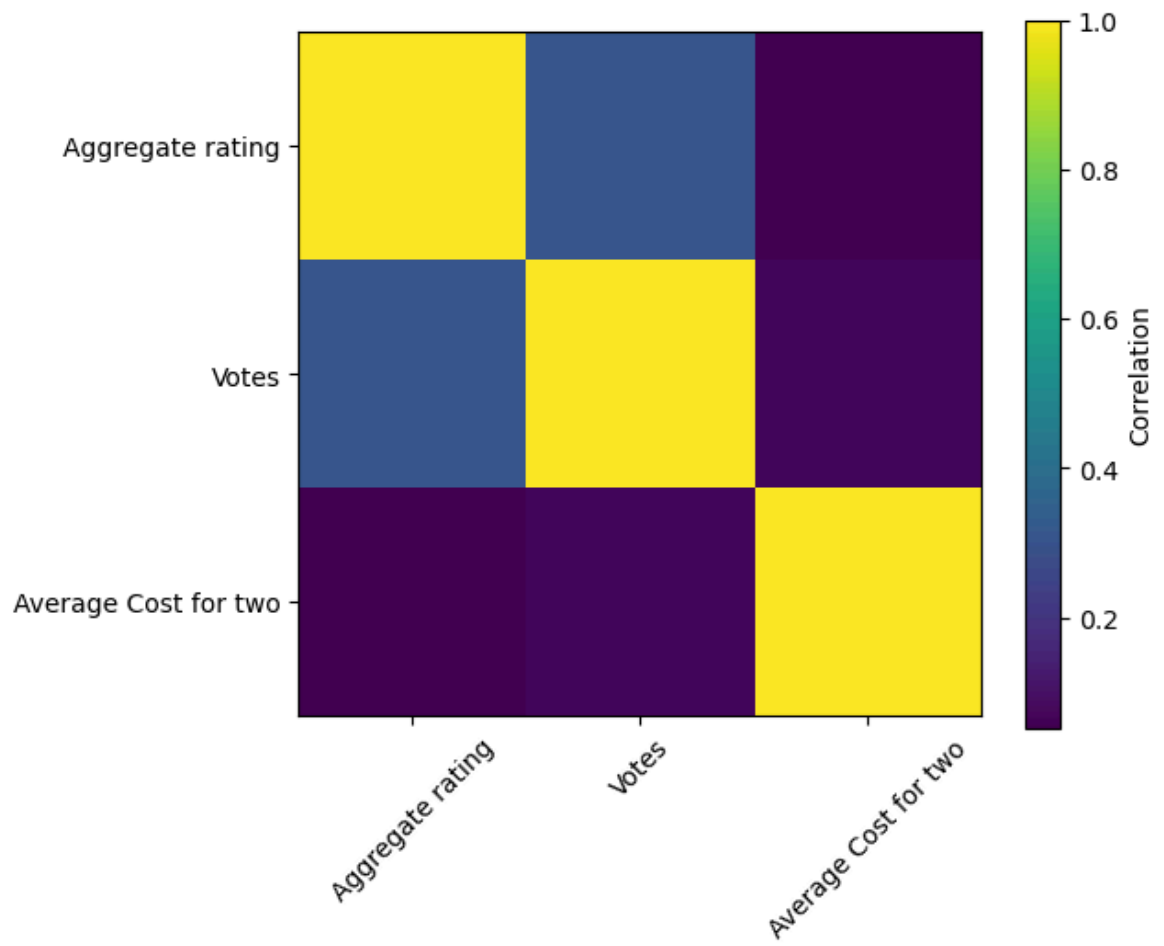
```
In [37]: import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
```

```
In [38]: df_corr = data[["Aggregate rating", "Votes", "Average Cost for two"]]
```

```
In [39]: corr = df_corr.corr()
```

```
In [41]: plt.figure(figsize=(6, 5))
plt.imshow(corr, cmap='viridis', interpolation='nearest')
plt.colorbar(label='Correlation')
plt.xticks(np.arange(len(corr.columns)), corr.columns, rotation=45)
plt.yticks(np.arange(len(corr.columns)), corr.columns)
```

```
Out[41]: ([<matplotlib.axis.YTick at 0x1fe55e7afd0>,
<matplotlib.axis.YTick at 0x1fe56284690>,
<matplotlib.axis.YTick at 0x1fe56285d10>],
[Text(0, 0, 'Aggregate rating'),
Text(0, 1, 'Votes'),
Text(0, 2, 'Average Cost for two')])
```



In []: